

VIKAS MOURYA

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SUMMARY

Backend Software Engineer with hands-on experience building scalable AI and web applications using Python, Django, FastAPI, and LangChain. Proven track record in deploying Retrieval-Augmented Generation (RAG) systems, automating workflows, and optimizing backend performance. Strong academic foundation (9.73 CGPA), published research in Springer, and a history of success in national hackathons.

EDUCATION

University of Mumbai

Bachelor of Engineering in Information Technology | **CGPA: 9.73 / 10.0**

2022 – 2026

Mumbai, Maharashtra

Coursework: Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Software Engineering

SKILLS

Languages: Python, JavaScript, C++, C

Frameworks and Libraries: Django, FastAPI, Flask, LangChain, Streamlit, Next.js

Tools and Platforms: AWS (EC2, S3), Docker, Git, Vercel, Render, Swagger, CI/CD Pipelines, Postman

Databases: PostgreSQL, MySQL, NeonDB, Chroma (Vector Store)

Core Concepts: REST APIs, Microservices, RAG Systems, Semantic Search, Token Auth, OAuth 2.0, Unit Testing, Agile

PROJECTS

StudyRAG | Python, FastAPI, LangChain, Streamlit

[GitHub](#)

- Engineered a RAG-based AI system for academic content Q&A using uploaded files (PDFs, PPTs, JSON), reducing manual note searching by over 80%.
- Integrated FastAPI backend with Chroma vector database and Streamlit frontend for seamless semantic search and document handling.
- Leveraged Gemini 1.5 Pro and Google Gen AI for reference-backed responses, improving study efficiency for 50+ users by up to 60%.

Morpheus | Python, Django, DRF

[GitHub](#)

- Built a Google Forms-style web app with 100+ customizable question types, drag-and-drop interface, and real-time data sync.
- Developed analytics modules to process 500+ form responses and automate result aggregation.
- Integrated secure REST APIs with token auth, QR sharing, and Swagger documentation for developer usability.

EXPERIENCE

Vighnotech

July 2024 – January 2025

Backend Developer

In-office

- Integrated intelligent automation into IMS and CRM systems, reducing workflow redundancy by 30%.
- Deployed scalable Retrieval-Augmented Generation (RAG) pipelines using FastAPI, Scrapy, and ChromaDB for semantic enterprise search.
- Designed and optimized a multi-agent chatbot system, reducing backend latency and improving query routing efficiency.

Jurisphere

March 2024 – April 2024

Full Stack Developer

Remote

- Improved semantic retrieval accuracy by 25% by optimizing RAG pipeline curation and preprocessing.
- Developed 15+ robust Scrapy-based web scrapers, reducing manual data collection by 40%.
- Streamlined reporting through automated Python tools, enhancing team productivity by 35%.

AWARDS AND ACHIEVEMENTS

- 1st Prize in National Level Hackathon at IIT JODHPUR for developing an AI-powered solution.
- Co-authored a paper on "Unsupervised vs. Supervised RAG" accepted at IOBDM 2025, published by **Springer**.
- Solved 400+ DSA problems on LeetCode, CodeChef (2), HackerRank (5), and GFG.
- Finalist in 5+ national hackathons and technical events.